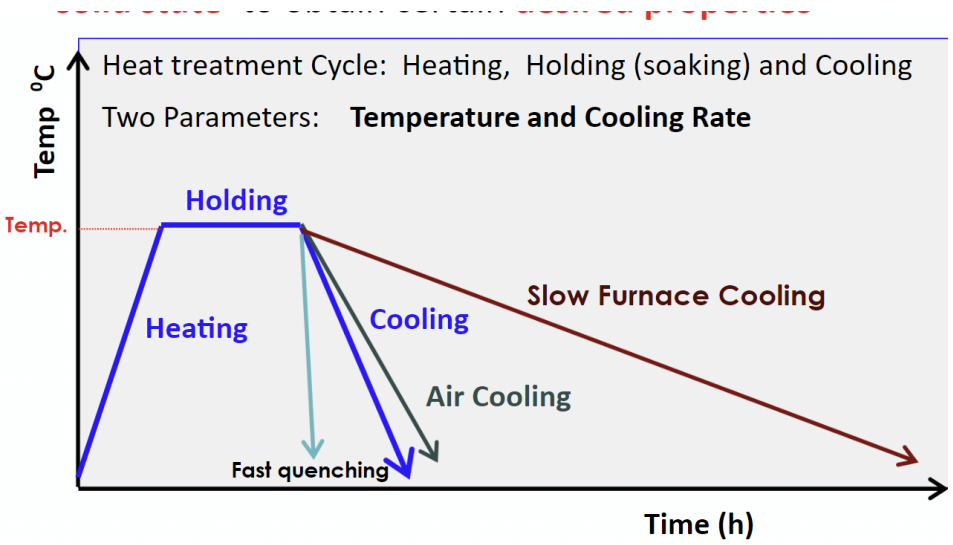
Heat Treatment

Heat treatment refers to heating and cooling a metal/alloy while solid to obtain/manipulate certain properties

The general heat treatment process is heating the metal in a furnace to a certain temperature(Heating), holding it at said temperature(holding/soaking), before cooling it down(cooling)

The two parameters that can be modified are the temperature (the temperature it is heated to and held at) and the cooling rate.

The cooling rate can be modified by cooling it differently, like leaving it to cool in the furnace(slowest), air cooling outside of the furnace, or fast quenching, which is dunking the metal in a cooling medium like water or oil(fastest).



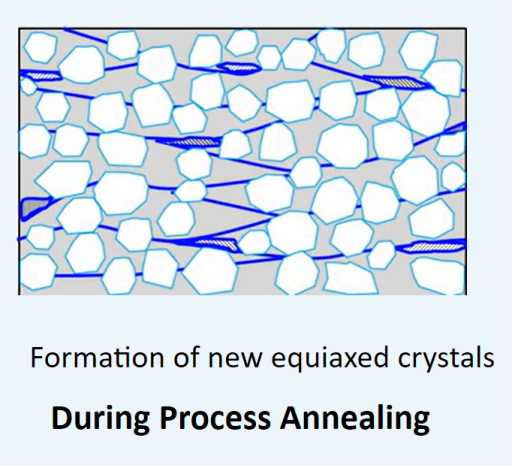
There are 4 main heat treatment processes: Annealing, Normalising, Hardening and Tempering.

Annealing is cooled in a furnace, hardening is cooled by quenching, and normalising and tempering are cooled by air.

Annealing

- As said, annealing is heating a metal in a furnace and is held at a fixed temperature, before turning off the power to leave the metal to cool in the furnace
- Recall that cold working causes the grains to be distorted, and that cooling metal in a mould also creates non uniform grains(creates fine grains and columnar grains at the boundaries).
- Annealing helps to reverse the distorted grains of cold working by forming back equiaxed grains, which helps to soften back metals and relieve internal stresses due to cold working
- Annealing also makes the grains uniform in metals cooled in moulds, making them easier to work with.
- There are 3 types of annealing, process annealing, spheroidising annealing and full annealing
- Process Annealing
  - Process Annealing is carried out either during or after cold working.
  - Process annealing helps to restore ductility and relieve internal stresses in the metal, which helps to allow cold worked parts to undergo cold working again without fracturing
  - For steels, steels with 0.4%C have high ductility, making it suitable to undergo process annealing.
  - Process annealing is carried out by heating the steel to 550C to 700C, where it is held for 1h before cooling in the furnace. Notice how this is below LCT.
  - Originally, the grain structure of a cold worked steel has distorted grains. Heating to 550 to 700C alloys the atoms in the metal to diffuse throughout the metal, as the temperature is above the activation energy needed to diffuse the atoms. As the atoms diffuse throughout the metal, new nuclei and the grains start to form and recrystallise, replacing the distorted grains with equiaxed grains, thus forming equiaxed grains in the end.

This makes the metal have the same original microstructure before cold working, making it more ductile and soft.



Spheroidising Annealing

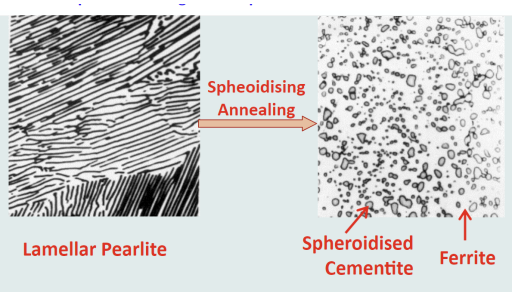
Spheroidising annealing is used to improve the machinability and ductility of high carbon and carbon tool steels.

Thus, steels with 0.5%C are better suited for this, as they are usually hard.

Spheroidising Annealing is heating the steel to 650C to 700C, just below LCT, before being held for 24 hours, then cooled in the furnace.

Consider the pearlite in steels with >0.5%C.

Originally, pearlite consists of layers of cementite and ferrite, however under heating below LCT, the cementite layers break up into small particles, forming into spherical lumps of cementite. The final microstructure is now spheroidised cementite in a ferrite matrix.



Now, when these spheroidised annealed steels are machined, instead of having to cut through the hard layers of cementite, the tool bit can slide between the spheroidised cementite, making it easier to machine. The spheroidised cementite can also slide past one another under deformation, making them more ductile

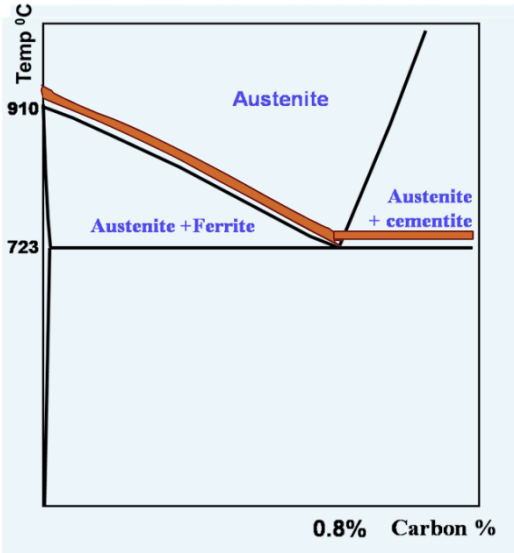
Full Annealing

Full Annealing is used to completely soften steel and to unify the non-uniform grain structure to make all the grain equiaxed.

Usually carried out before machining, cold working, other machining/heat treatment processes, to prepare the material for such processes.

Full Annealing can be carried out on steel with any carbon content

For hypo-eutectoid steel, the steel is heated to 30 to 50C above UCT, while for hyper-eutectoid steel, the steel is heated to 30 to 50C above LCT.



The heating temperature depends on the carbon content of the metal.

The steel is held sufficiently so that phase transition can take place, where either austenite or austenite + cementite is formed

The steel is then cooled in the furnace to allow the microstructure to form back into the room temp microstructure, however now with equiaxed grains.

As the microstructure has equiaxed grains instead of columnar grains, which make the metal harder and less ductile, the treated steel is now softer.

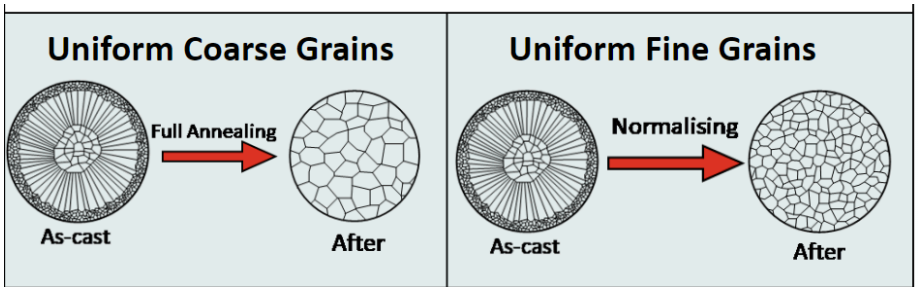
Normalising

Similar to full annealing, normalising also obtains a uniform and fine grain structure, and to relieve stress between machining.

Normalising applies for all steels with any carbon content.

Normalising is heating the steel 30 to 50 C above the UCT regardless of %C, then holding it to allow the microstructure to change to austenite. The metal is then cooled in still air.

The final grain structure is a uniform and FINE grain structure due to the faster cooling rate, making it stronger and harder.



Note the different in the grain density, and the final mechanical properties of the treated metal.

Hardening

hardening is used to increase the hardness, wear resistance and strength of a material.

Steels with high carbon content are used so that their high hardness can be further increased.

Similar to full annealing, the process starts with heating to 30 to 50C above UCT, while for hyper-eutectoid steel, the steel is heated to 30 to 50C above LCT, similar to full annealing. The steel is then held sufficiently for austenite to form, before being quenched in oil.

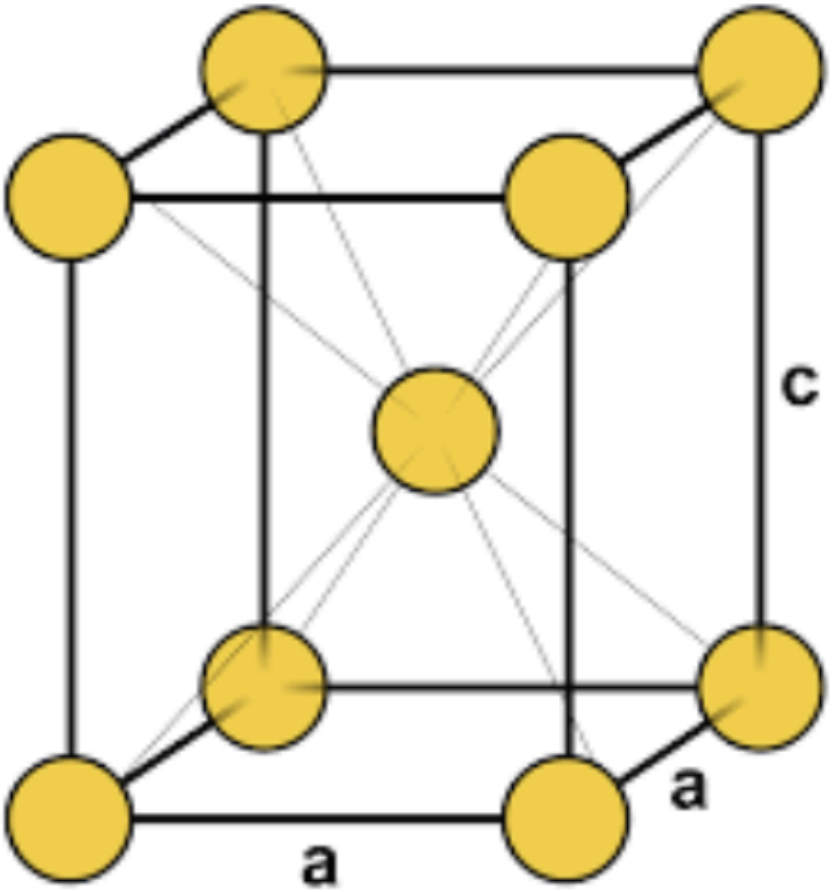
After hardening, Martensite is formed, a very hard steel microstructure. Note that the only difference between hardening and full annealing is the cooling, where this is quenched in oil.

Under normal slow cooling from above the UCT/LCT, carbon is able to diffuse throughout the steel, allowing the phases to change as the time is enough to allow the large amount of carbon to diffuse throughout.

With fast cooling, the carbon IN PEARLITE is unable to diffuse throughout the metal for the phase transformations due to too little time.

This causes a large amount of atoms to have a slight displacement, causing a distortion in the lattice and creating a new crystal structure called Body-Centered Tetragonal (BCT lattice). This microstructure (for steel) is called Martensite. Martensite is hard and brittle, with a needle shaped microstructure. Martensite does not have any grains.

In general, fast cooling of pearlite turns it into martensite, with other microstructures like ferrite and cementite unaffected.



The BCT and BCC strcuture is different in that the BCT box is taller (lol)

Critical Cooling Rate & Hardenability

For all steels under heat treatment, there exists a slowest possible rate of cooling during quenching to produce martensite/martensitic structure. This rate is called the critical cooling rate, or CCR.

Generally, a high carbon steel/alloy has a slow CCR.

Thus, another property, hardenability, the ease with which a material/steel can be hardened, can be derived.

If a metal has a slower CCR, it is easier to cool it down faster than that rate compared to a faster CCR. Thus, a slower CCR will be easier to form martensite, thus making it easier to be hardened.

Thus,

high carbon steel/alloy ⟶ slower CCR ⟶ high hardenability

In summary, hardening requires 3 conditions:

Carbon content of steel > 0.3%

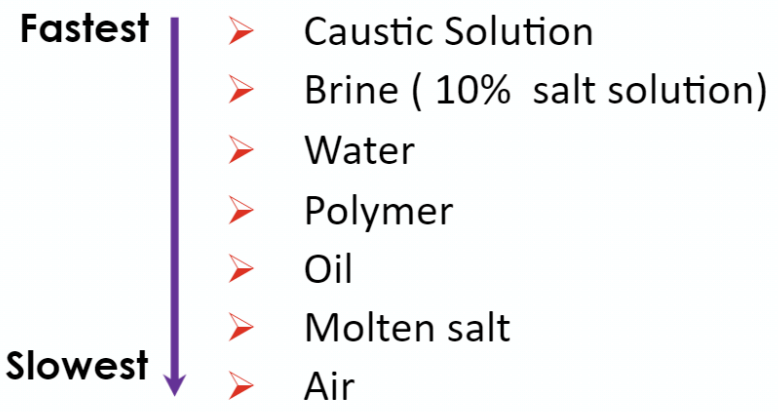
Heated and held 30 to 50 C above UCT/

LCT for austenite to form

Cooled faster than CCR

Quenching Media

Quenching media refers to the substance the metal is cooled in.



Different cooling mediums have different cooling rates. Caustic solutions refers to a NaOH or KOH solution with 10% of the hydroxide in water. This solution is able to absorb the most amount of heat the fastest, thus causing the fastest cooling rate.

As a rule of thumb:

Water is used for (low to medium carbon) plain carbon steels (less carbon ⟶ higher CCR)

Oil is used for high carbon plain carbon steels and alloy steels (more carbon/alloy steel ⟶ lower CCR)

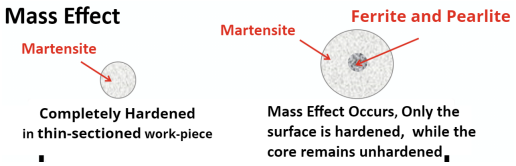
Air blast quenching are usually for high speed alloy steels (high alloy steels ⟶ lowest CCR)

Mass Effect

Mass effect refers to how the cooling rate of the outside and inside of the steel cools at different rates; the center cooling slower than the surface(thus causing only the surface to be hardened).

This causes the surface to have a martensite microstructure, while the core has a microstructure of ferrite and pearlite

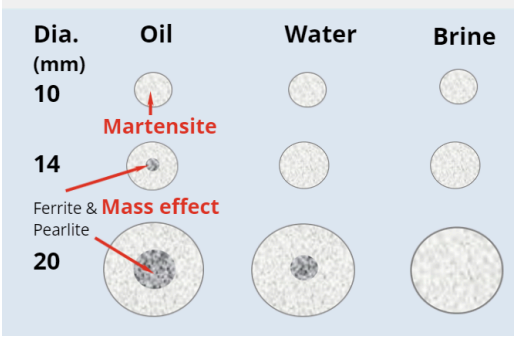
If the material is too thick or if the cooling medium cannot cool the metal fast enough, the center of the metal may not form into martensite.



This may cause potential distortion and cracking, due to the different cooling rates causing different contractions, or due to the microstructures forming at different times.

To avoid mass effect, a proper quenching medium should be picked.

The quenching medium should be fast enough to exceed CCR and avoid mass effect, but to be slow as possible to minimise distortion and cracking.



This helps to show the quenching medium appropriate for each thickness.

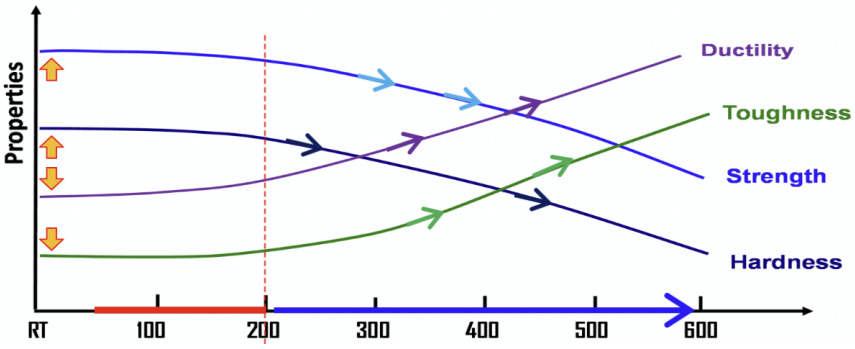
However, if all quenching media will work, oil is the most appropriate.

Tempering is carried out on steels after hardening, to improve the toughness and ductility, and to remove internal stresses due to quenching.

Tempering

Tempering works by reheating the steel to below the LCT, before being held sufficiently long enough and cooled in still air.

The tempering temperature the steel is held at influences its properties.



Recall that Martensite is very strong and hard, thus it will not be very ductile or tough.

Considering this, these properties will not (significantly) change when heated up to 200C. However heating up to 200C helps to relieve quenching stresses, whilst keeping its properties.

Past 200C, the martensite is tempered, where its strength decreases.

Its hardness will also decrease with it, and the ductility and toughness will increase (opposite to strength).

Overall, tempering helps to improve the toughness and ductility while keeping strength and hardness, making it useful for crankshafts and axles.

The table summarises all of the heat treatment processes:

	Process Ann.	Spheroidising Ann.	Full Ann.	Normalising	Hardening	Tempering (remove stress)	Temperi (add duc
Purpose	restore ductility, relieve internal stresses for cold worked parts	improve machinability of high carbon/carbon tool steels	Soften steel, form back uniform grain structure	form back uniform grain structure, relieve stress between machining	To improve strength, hardness and wear resistance.	To remove internal stresses due to quenching after hardening	To impro ductility toughne hardenin
Steel Type	~0.4%C	>0.5%C	any	any	>0.3%C	hardened steel	hardene
Heating Temp	550C to 700C	650C to 700C	30C to 50C above UCT for hypo, 30 to 50C above LCT for hyper	30C to 50C above UCT	30C to 50C above UCT for hypo, 30 to 50C above LCT for hyper	<200C	>200C
Holding Time	1h	24h	sufficient/until austenite	sufficient/until austenite	sufficient/until austenite	sufficient	sufficien
Cooling medium	Slow Furnace Cooling	Slow furnace Cooling	Slow furnace Cooling	Air cooled	Quenched in oil	Air cooled	Air coole
Microstructure	Distorted grains → equiaxed grains	Laminar Pearlite → spheroidised cementite in ferrite lattice	Non-uniform grains → uniform coarse grains	Non-uniform grains → uniform fine grains	Austenite in steel→ Martensite	Remains as martensite	Martensi tempere martensi
Theory/Reason	the heating allows atoms to diffuse in the steel → grains can recrystallise → equiaxed grains replace distorted grains	cementite breaks up into small particles under heating → small spheres of cementite can slide past tool bit/one another → ductile	grains are recrystallised → equiaxed grains are formed ⟶ slower cooling rate forms coarse grains	grains are recrystallised → equiaxed grains are formed ⟶ faster cooling rate forms finer grains	quick cooling rate prevents carbon from diffusing, atoms are displaced slightly → austenite forms into martensite, a hard and strong microstructure with BCT crystal structure	heating below 200C does not significantly change microstructure, but relieves internal stresses	heating a 200C re the hardı and strer while increasir toughne ductility, making t final stee having a mix of st and toug